

Algorithmic Pseudocode Sample 24

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

```
1: Input: Stock data for two assets  $S_1$  and  $S_2$ , buy threshold, sell threshold
2: Output: Trade signals for pairs trading
3:
4: procedure COMPUTE HEDGE RATIO(data1, data2)
5:   model  $\leftarrow$  perform OLS regression (data1, data2)
6:   return model.params[1]
7: end procedure
8:
9: procedure INITIALIZE(fast, slow)
10:   hedge_ratio  $\leftarrow$  COMPUTE HEDGE RATIO( $S_1$ ,  $S_2$ )
11:   Calculate spread  $\leftarrow S_1 - \text{hedge\_ratio} \times S_2$ 
12:   fast_ema  $\leftarrow$  EMA(spread, fast)
13:   slow_ema  $\leftarrow$  EMA(spread, slow)
14:   apo_spread  $\leftarrow$  fast_ema - slow_ema
15:   position  $\leftarrow$  0
16: end procedure
17:
18: procedure NEXT(buy threshold, sell threshold)
19:   if apo_spread < buy_threshold then
20:     Execute Buy for  $S_1$  and Sell Short for  $S_2$ 
21:   else if apo_spread > sell_threshold then
22:     Execute Sell Short for  $S_1$  and Buy for  $S_2$ 
23:   end if
24: end procedure
```
